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Modular Framework for Automated QSM Integration into Deep Brain Stimulation Neurosurgical Planning Systems

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QUANTITATIVE SUSCEPTIBILITY MAPPING SOFTWARE TOOLS DEEP BRAIN STIMULATION CEREBELLUM AND DENTATE NUCLEUS NEUROSURGICAL PLANNING

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Impact

An open-source, modular tool for automated clinical integration of QSM can enhance surgical planning for deep brain stimulation and other neuro applications. The framework can be adapted to any non-conventional image contrast, accelerating clinical translation of advanced MRI.

Synopsis

Motivation: Precise lead placement is critical for effective deep brain stimulation and depends on preoperative imaging quality. Quantitative susceptibility mapping (QSM) improves target visualization, but clinical adoption remains limited.
Goals: Make QSM accessible for neurosurgical planning by developing software that overcomes processing and compatibility barriers.
Approach: We developed a framework that retrieves DICOMs from clinical PACS, automatically generates QSM with anatomical alignment, and outputs PACS-compatible DICOMs for neurosurgical planning.
Results: The framework was tested on GE and Philips DICOMs, demonstrating compatibility with StealthStation and Brainlab. Run time and registration error ranged 216s–486s and 0.80mm–1.73mm across five patients.

Introduction

The efficacy of deep brain stimulation relies on precise lead placement within the intended anatomical target, which is directly influenced by the quality of preoperative imaging. Iron-sensitive quantitative susceptibility mapping (QSM) offers a superior contrast-to-noise ratio for enhanced visualization of target deep gray matter structures, demonstrating more accurate delineations,[1–3] and a more intuitive and simplified approach[4].

Despite interest in QSM for DBS neurosurgical planning, several drawbacks have hindered its clinical adoption. This includes sequence implementation challenges which can be addressed with support from institutional and/or manufacturer-affiliated MRI scientists and the use of recommended parameters[1,5]. An less resolved challenge is that major scanner manufacturers lack routine on-scanner QSM processing capabilities, forcing reliance on third-party software[6,7] and clinical team expertise. QSM images are typically output in NIfTI rather than DICOM formats, rendering them incompatible with clinical PACS for straightforward access in surgical planning software.

To make QSM imaging more accessible for neurosurgical planning, we developed an automated framework that retrieves DICOM files from clinical PACS, generates a QSM image aligned to an anatomical reference, and outputs QSM DICOMs compatible with PACS and neurosurgical planning software.

Methods

We developed an end-to-end MRI framework made up of (1) Input, (2) Processing, (3) DICOM Generation, and (4) Output stages (**Fig 1**). This modular set up allows any stage to be replaced, depending on user needs and institutional IT infrastructure.

Input: This module receives and organizes DICOM files for processing. It accepts directories containing T2*-weighted magnitude and phase (or real and imaginary) DICOMs to generate the QSM image, along with DICOMs for an FGATIR[8] (or T1-weighted) sequence, to serve as an anatomical reference for QSM image alignment. Depending on institutional infrastructure, the input DICOMs can be retrieved automatically or manually from clinical scanners or a web-based PACS. Saving the DICOMs automatically triggers the Processing module through a user-configurable listener mechanism.

Processing: This module produces a QSM image aligned to the supplied FGATIR image, in NifTI format. Brain extraction and masking are initially performed using HD-BET[9] to mitigate processing artifacts. The skull stripped images are then fed into the end-to-end QSMxT pipeline (v8.1.3)[7] to produce the QSM image contrast; however, QSMxT can easily be interchanged with any preferred pipeline. Finally, the FGATIR image and the unprocessed T2*-weighted magnitude image are registered using ANTs[10] (necessary for neurosurgical planning), and the calculated transform is applied to the processed QSM image to produce an FGATIR-aligned QSM image.

DICOM Generation: This module uses a custom executable to convert the FGATIR-aligned QSM NifTI image into DICOM files compatible with both Medtronic's StealthStation (model S8, version 1.0.3.) and Brainlab Elements (v6.0) neurosurgical planning software. It leverages the single 3D volume DICOM format of the provided FGATIR image and assigns a random Series Instance UID to ensure unique identification when re-integrated with other images from the same scanning session. Essential for surgical planning, the module ensures QSM DICOMs orientation correctness by integrating novel, custom software called 'Flip Proof' that leverages the compiler to enforce orientation-safe image operations.[11]

Output: This final module automatically transmits the orientation-safe, FGATIR-aligned QSM DICOMs to clinical PACS using an institution-specific IT solution from where they can be uploaded to neurosurgical planning software. A manual solution also exists.

Performance Assessment: The framework was tested using DICOMs from different scanner manufacturer (GE, Philips; TR/TE/delta/TE=37ms/5ms/9.5ms echoes=3 or 4, FOV=220x220mm, 1x1x1.2mm voxels, flip=15, parallel imaging), and compatibility with StealthStation and Brainlab planning software was evaluated. Run time and target registration errors between FGATIR and QSM were measured for five patient datasets (with consent and IRB approval), with errors quantified as the Euclidian distance between corresponding manual landmarks on the globus pallidus internus and cerebellar dentate nucleus.

Results

The end-to-end framework was successfully run. QSM DICOMs enable superior visualization of target structures (Fig 2) and were compatible with clinical PACS, StealthStation, and Brainlab (Fig 3). Run time ranged 216s–486s, primarily due to the Processing module, while registration errors ranged 0.80mm–1.73mm (Fig 4, Fig 5).

Discussion

Our simple, modular imaging framework eliminates key barriers to QSM adoption for DBS surgical planning with notable features including a user-configurable listener mechanism, automated QSM processing with FGATIR alignment, and PACS-compatible DICOM generation with preserved orientation. The modular design accommodates different institutional infrastructures, while compatibility with multiple MRI manufacturers and surgical planning software ensures broad applicability. Ongoing work will further test the framework's robustness against heterogeneous inputs (e.g., partial datasets), integrate additional QSM processing methods, and deploy an initial version on GitHub (Radiology-Morrison-lab-UCSF/modular-image-processing-system).

Acknowledgements

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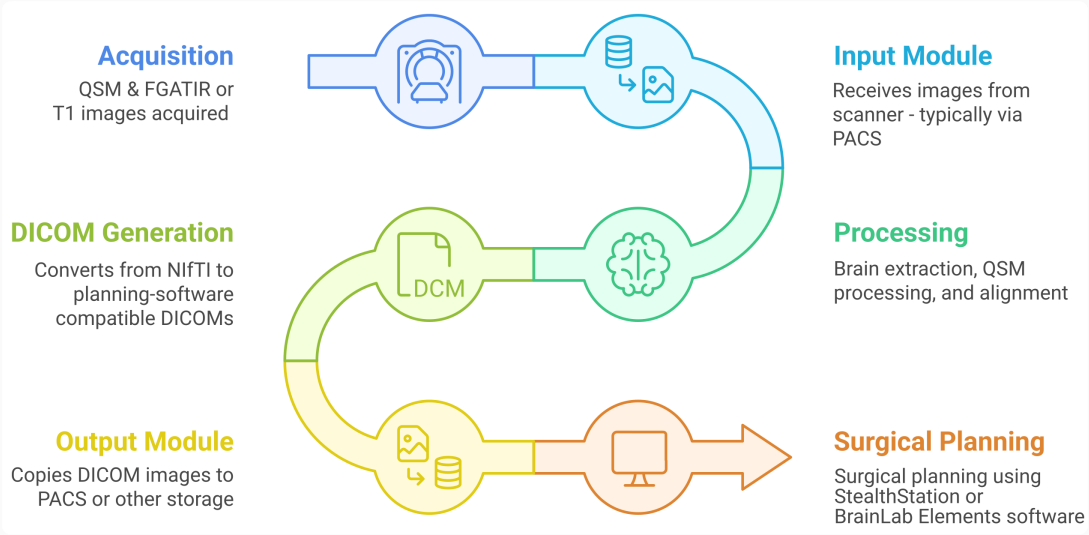
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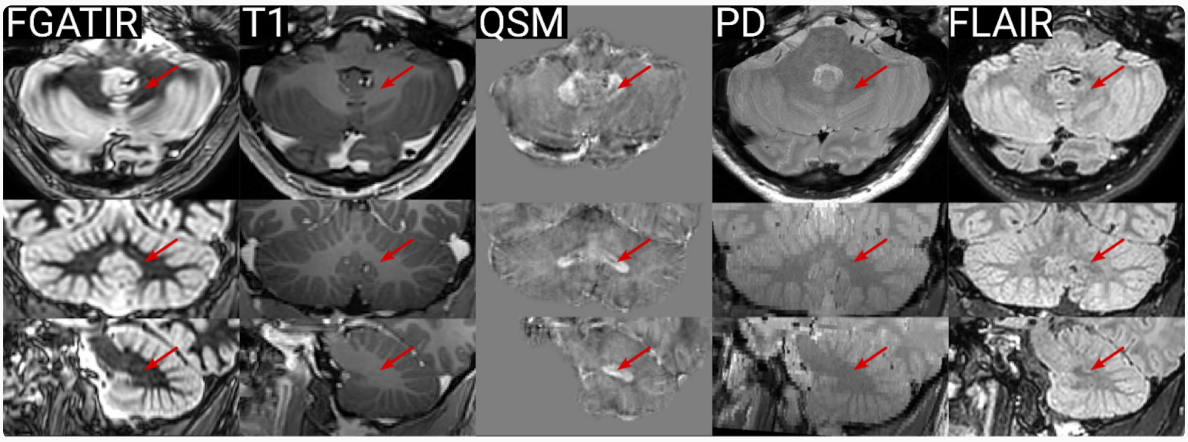
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Figures and Tables



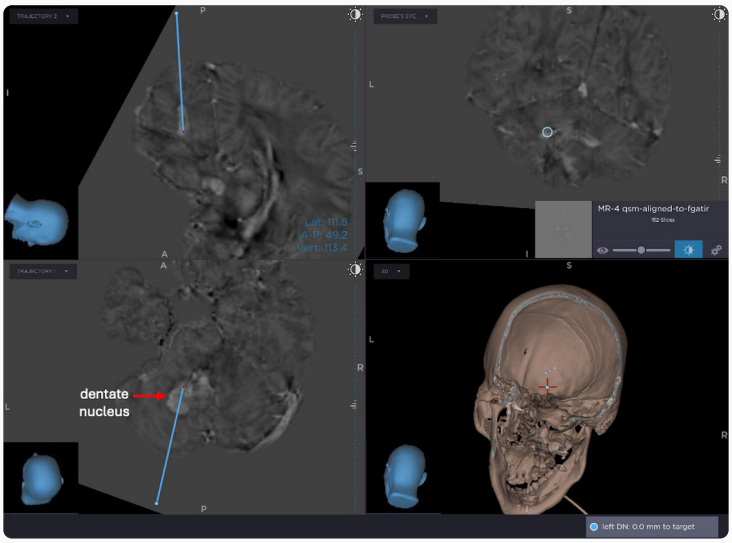
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Figure 1: *Schematic of the modular imaging framework for quantitative susceptibility mapping (QSM) integration into neurosurgical planning workflows. Input and Output modules can be replaced by institution-specific information technology solutions. The user can also replace the QSM processing algorithm within the Processing module. If a PACS solution does not exist, the user can move data via a memory stick or network drive, as appropriate.*



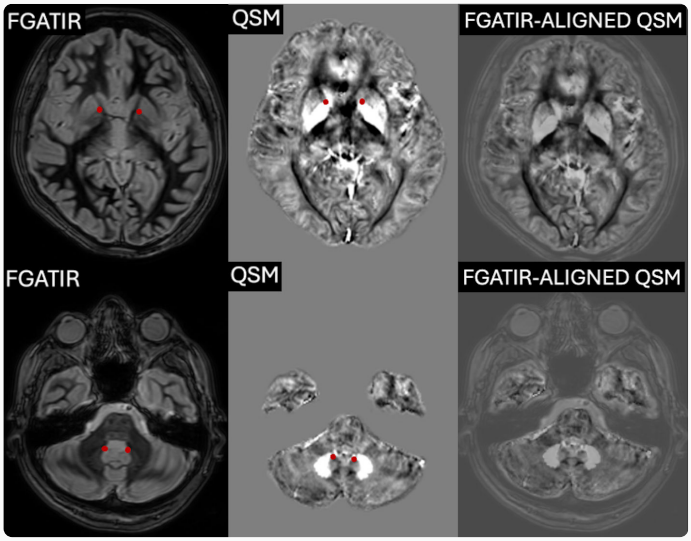
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Figure 2: *Representative patient case demonstrating QSM's superior visualization of the cerebellar dentate nucleus, a target for dyskinetic cerebral palsy. Arrows indicate the left dentate nucleus at the level of displayed slices. Only the processed QSM image provided sufficient tissue contrast to confidently identify the borders of the dentate nucleus.*



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Figure 3: *Representative patient case illustrating FGATIR-aligned QSM DICOM compatibility with Medtronic's StealthStation. In this example, the QSM image was used to guide targeting of the dentate nucleus of the cerebellum for deep brain stimulation treatment of dyskinetic cerebral palsy (blue line is the planned lead trajectory). QSM DICOMs can be visualized independently, as shown here, or in conjunction with FGATIR and/or other conventional images to better delineate the skull-tissue boundary.*



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Figure 4: *Registration results for a representative patient. The native FGATIR and native QSM images are shown with the FGATIR-aligned QSM image overlaid onto the native FGATIR image. The processing module achieved good alignment while preserving the QSM iron-sensitive contrast. Slice levels highlight the globus pallidus internus (top row) and cerebellar dentate nucleus (bottom row) within which landmarks were positioned to calculate target registration error (red dots provide an illustrative example).*

Summary of Performance Measures

Subject	Scanner Manufacturer	Raw QSM Format	Processing Run Time (s)	DICOM Generation Run Time (s)	Error L_2 (mm)
1	Philips	Phase/Mag	452	2	0.80
2	Philips	Phase/Mag	243	3	1.33
3	Philips	Phase/Mag	212	4	1.55
4	General Electric	Real/Imag	410	2	1.33
5	General Electric	Real/Imag	484	2	1.73

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Figure 5: *Summary of framework run time and target registration error evaluated across five patient datasets during pilot testing.*